ML-KEM CBD RTL optimization on VTR

Prospectively frozen paired PPA estimate · open 45 nm FPGA proxy

A fixed-chunk state representation reduces the paired area-delay-power estimate by 9.7338%

Prepared by Göther Labs

Measurement completed: 21 August 2026

1. Executive result

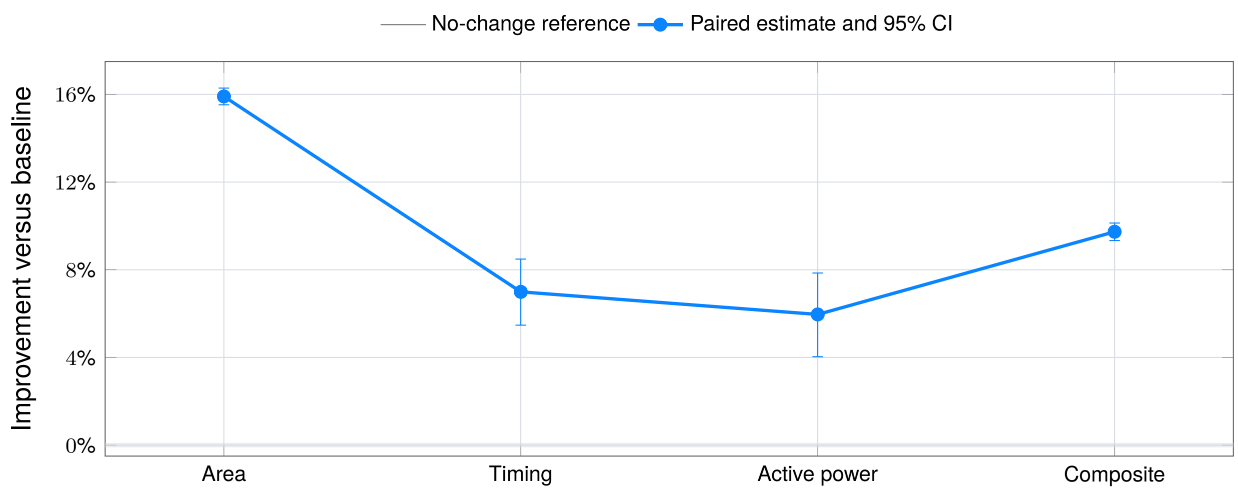
FROZEN BASELINE VERSUS OPTIMIZED RTL

The optimized RTL improves the equal-weight area-delay-active-power estimator by 9.7338% (95% CI: 9.3331% to 10.1327%) without changing the interface, defined-input cycle behaviour, latency, output order or ML-KEM CBD coefficient arithmetic.

Primary metric	Baseline aggregate	Optimized aggregate	Paired improvement	95% CI / state
Total area	22,751,867 MWTA	19,132,428 MWTA	**15.9083%**	15.5265% to 16.2884% · **better**
Critical path	4.0538 ns	3.7703 ns	**6.9920%**	5.4724% to 8.4872% · **better**
Active total power	64.106 mW	60.284 mW	**5.9621%**	4.0336% to 7.8519% · **better**
Composite estimator	1.000000	0.902662	**9.7338%**	9.3331% to 10.1327% · **better**

Paired primary PPA improvement

64 prospectively frozen publication pairs · lower is better · bars are two-sided 95% confidence intervals



The aggregate values are geometric means across the fixed publication sample. The reported composite 0.902662 is the paired geometric-mean estimator; the median of the 64 per-pair composite ratios is 0.901744. The estimator and its predeclared paired interval define acceptance.

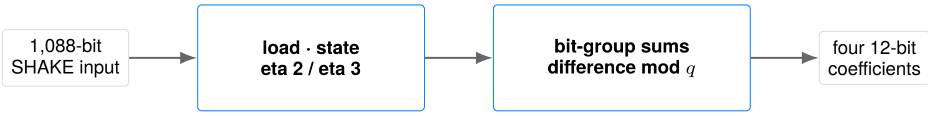
2. Module and benchmark

WHAT IS BEING OPTIMIZED

The evaluated module is the stateful centred-binomial-distribution (CBD) sampler from the public HOPE-MLKEM hardware implementation. The exact upstream source is `rtl/cbd.v` at pinned commit `72a90d80...`, under the MIT License.

ML-KEM, standardised in NIST FIPS 203, uses small sampled polynomial coefficients as controlled mathematical noise. This block consumes 1,088 pseudorandom SHAKE bits, supports the eta-2 and eta-3 sampling modes, and emits four 12-bit coefficients per active cycle through two 24-bit write words. It is one synthesizable cryptographic building block, not a complete key-encapsulation engine.

```
module CBD(clk, rst, load, start, eta, scnd, in_shake,
           end_op, en_write, data_in_1, data_in_2,
           addr_1, addr_2);
  input clk, rst, load, start, scnd;
  input  [1:0] eta;        input  [1087:0] in_shake;
  output end_op, en_write;
  output [23:0] data_in_1, data_in_2;
  output [7:0] addr_1, addr_2;
endmodule
```



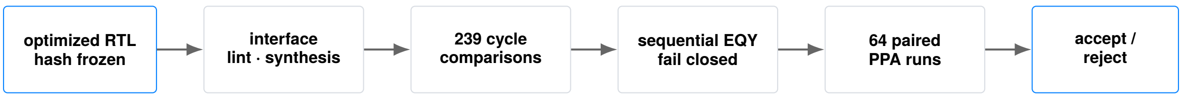
Only `cbd.v` is editable. Reset, load and second-load semantics, stalls, eta modes, write enable, addresses, termination, outputs and all defined-input cycle timing remain fixed.

Artifact	SHA-256
Frozen baseline `cbd.v`	`c152f96a682ea390c610dc32ee00a317c0ad990f9b082451325cdaf12f2218b0`
Optimized `cbd.v`	`da0b4e79beb9cf9cc24c3ce949068c3a29316273a6320e283f4201fbd884c9ef`

3. Evaluation contract

FROZEN BEFORE PUBLICATION MEASUREMENT

Dimension	Frozen choice
Editable artifact	`cbd.v` only; baseline and optimized sources frozen by SHA-256.
Functional behaviour	Exact interface, reset, loading, modes, stalls, addresses, termination and defined-input cycle outputs.
Golden simulation	239 deterministic dual-design cycles and 239 comparisons.
Formal semantics	Two-state sequential equivalence after two active-low reset cycles; full 1,096-bit joined logical-state refinement.
Formal tool	EQY `v0.67-1-g6734d8c`; SAT five-step temporal induction; fail closed.
Physical flow	VTR/VPR commit `95f5c6de...`; pinned Linux/amd64 image; runtime network disabled.
Architecture	`k6_N10_I40_Fi6_L4_frac0_ff1_45nm.xml`; homogeneous LUT6 target.
Power	VTR PTM45, 0.9 V, $85\,^{\circ}$C; deterministic ACE probabilistic activity.
Search sample	Seeds 1, 7 and 19; optimizer-visible provisional ranking only.
Initial confirmation	12 fixed disjoint pairs; observed before the publication protocol.
Publication sample	Seeds 101 through 164; disjoint, fixed $n = 64$, no extension.
Primary score	Equal-weight paired area, delay and active-total-power geometric mean.



Exact decision rule. All correctness and evidence-integrity gates must pass. The composite one-sided 95% upper log-ratio bound must be below zero, and no primary metric may have a one-sided 95% lower log-ratio bound above zero. Otherwise the result is inconclusive or regressed.

The publication protocol was frozen before either publication baseline or optimized measurement. The prior 12-pair result, with 9.4652% improvement, is disclosed as selection history and is not pooled into the 64-pair confirmation.

4. Verification before PPA

FAIL-CLOSED EVIDENCE FOR THE OPTIMIZED HASH

Gate	Published evidence	Result
Structural policy	Exact module, interface, hierarchy, patch and synthesizable Verilog identities	**PASS**
Cycle regression	Reset, eta 2/3, stalls, reload, second-load, completion and cycle outputs	**PASS**
Sequential formal	Five-step temporal induction over the complete 1,096-bit logical-state relation	**PASS**
Physical validity	Route success, finite metrics, resource identities and complete paired extraction	**PASS**
Raw provenance	All 541 archive members and all 128 post-route PPA rows rechecked	**PASS**

The cycle regression compares the frozen baseline and optimized design at every observed cycle, not only at final completion. It exercises reset, both supported eta modes, active and stalled cycles, first and second loads, address advancement, write data and termination behaviour. A mismatch stops evaluation before PPA scoring.

The formal proof joins the two different physical representations through the logical `reg_shake` view. Five-step induction is not a five-cycle bounded test: it establishes the transition invariant for arbitrary future cycles after reset. Outputs remain proof obligations, while arithmetic temporaries are excluded as accidental match points. Inconclusive or timed-out proof is invalid.

```
python3 verify.py
```

> **Evidence boundary.** The compact check establishes certificate consistency and exact published identities. With the separately downloaded raw archive, the same verifier checks every member, correctness pass logs and all 128 baseline/optimized post-route rows. It re-extracts recorded metrics; it does not rerun Verilator, Yosys, EQY or VTR.

Cycle-test SHA-256: `3cf3c7d63a8a0ac14660e9436c9b860efd1a3efc5af595220639353cdf7bfeb1`

Formal-config SHA-256: `1ec06bad2ca3c1861f99c0615bdd6d83e2d09ca6b243ed9c7b665da657b83ee7`

5. Exact baseline-to-optimized diff

ALL CHANGED LINES FROM CBD.V

The patch below is generated directly from the two checked-in RTL files. No testbench, formal harness, architecture, activity model, metric parser, seed pool or tool flag is editable by the optimization candidate.

```
--- rtl/baseline/cbd.v
+++ rtl/optimized/cbd.v
@@ -21,19 +21,43 @@
     output [7:0]    addr_2
     );

+    // Keep the wide physical state on a fixed 24-bit shift. A two-bit byte
+    // phase represents the 16-bit eta=2 advance without a 16/24 mux on every
+    // flip-flop. reg_shake is the synthesized logical refinement view used by
+    // the datapath and by sequential equivalence.
+    reg [1095:0] shake_chunks;
+    reg [1:0] shake_phase;
     reg [1095:0] reg_shake;
+    always @* begin
+        case (shake_phase)
+            2'd0: reg_shake = shake_chunks;
+            2'd1: reg_shake = shake_chunks >> 8;
+            2'd2: reg_shake = shake_chunks >> 16;
+            default: reg_shake = shake_chunks >> 24;
+        endcase
+    end
+
     always @(posedge clk) begin
-        if(!rst) reg_shake <= 0;
+        if(!rst) begin
+            shake_chunks <= 0;
+            shake_phase <= 0;
+        end
         else begin
             if(load) begin
-                if(scnd)                    reg_shake <= {in_shake, reg_shake[7:0]};
-                else                        reg_shake <= {8'h00, in_shake};
+                if(scnd)                    shake_chunks <= {in_shake, reg_shake[7:0]};
+                else                        shake_chunks <= {8'h00, in_shake};
+                shake_phase <= 0;
             end
             else if(start & !end_op) begin
-                if(eta == 2'b11)            reg_shake <= reg_shake >> 24;
-                else                        reg_shake <= reg_shake >> 16;
+                if(eta == 2'b11) begin
+                    shake_chunks <= shake_chunks >> 24;
+                end
+                else begin
+                    if(shake_phase != 0)     shake_chunks <= shake_chunks >> 24;
+                    if(shake_phase == 0)     shake_phase <= 2;
+                    else                    shake_phase <= shake_phase - 1'b1;
+                end
             end
-            else                            reg_shake <= reg_shake;
         end
     end
```

> **Attribution boundary.** The measured PPA result belongs to this one complete accepted patch. The following pages are three overlapping structural views of the same patch, not three independently measured rewrites; no exact fraction of the gain is assigned to one statement.

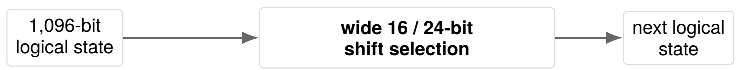
Patch SHA-256: `18ac09ca7be4c10484e5948d8798fee6fd1bfbe23b9f38989e18fc2a7da80b82`

6. Structural view 1: remove the wide variable shift

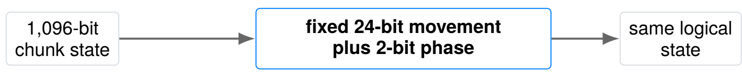
ONE ACCEPTED PATCH · SOURCE: OPTIMIZED CBD.V:24–62

The baseline stores the logical sampler state directly in 1,096 enabled flip-flops. Every active cycle asks synthesis to choose between a 16-bit and a 24-bit right shift according to `eta`. That variable movement appears around the complete wide state even though both modes consume the same data in a regular sequence.

Frozen baseline



Optimized RTL



The optimized RTL keeps the wide physical register in `shake_chunks`. Its only non-load movement is a fixed 24-bit shift. The two-bit `shake_phase` records an outstanding byte offset, and a combinational view presents the unchanged logical head to all original coefficient arithmetic.

If C_t is the physical chunk state and p_t is the byte phase, the logical state is

$$L_t = C_t \gg (8p_t), \qquad p_t \in \{0, 1, 2\}.$$

The datapath continues to read L_t exactly where it previously read the baseline state. The transformation changes representation, not the sampling formula.

> **Structural result.** Packed CLB demand falls from 337 to 270 (19.8813% fewer) under the fixed LUT6 architecture. This is an implementation result, not an RTL line-count estimate.

7. Structural view 2: encode eta-2 movement as phase

ONE ACCEPTED PATCH · SOURCE: OPTIMIZED CBD.V:31–38 AND 51–59

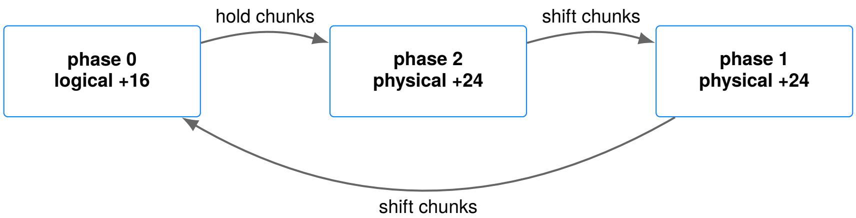
Eta-3 consumes 24 bits per active cycle, so the physical and logical movements already match. Eta-2 consumes 16 bits. The optimized representation realizes three 16-bit logical advances using two fixed 24-bit physical advances and the phase sequence $0 \to 2 \to 1 \to 0$.

Current phase	Physical action	Next phase	Logical advance
0	hold C_t	2	$0 + 16 = 16$ bits
2	$C_t \gg 24$	1	$24 - 8 = 16$ bits
1	$C_t \gg 24$	0	$24 - 8 = 16$ bits

For eta-2 the transition relation satisfies

$$L_{t+1} = C_{t+1} \gg (8p_{t+1}) = L_t \gg 16$$

in all three phase cases. For eta-3, the phase is retained while $C_{t+1} = C_t \gg 24$, so the same relation gives $L_{t+1} = L_t \gg 24$. This remains true even if defined input sequences change `eta`; the proof does not depend on a software promise that the mode stays constant.

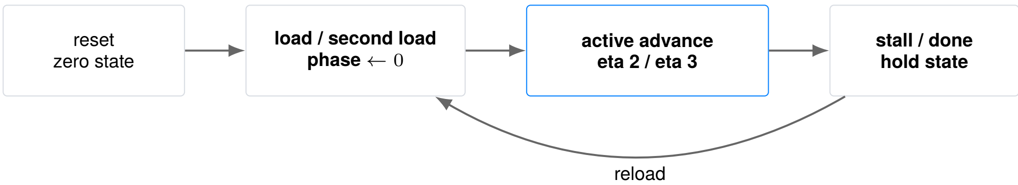


Scope. This representation is specific to a stream whose supported logical advances are 16 and 24 bits and whose load boundary restores phase zero. Different consumption widths or load semantics require a new invariant and a new evaluation.

8. Structural view 3: preserve loading, stalls and outputs

ONE ACCEPTED PATCH · SOURCE: OPTIMIZED CBD.V:40–62 AND 64–134

The optimization is safe only if representation metadata follows every control path. Reset clears both chunk state and phase. A first load writes `{8'h00,in_shake}` and restores phase zero. A second load preserves the low byte of the current logical view, not an arbitrary physical byte, then also restores phase zero.



When `start` is low or `end_op` is high, neither `shake_chunks` nor `shake_phase` is assigned, so both retain state. The original counter, coefficient equations, write enable, write addresses and completion logic are byte-for-byte unchanged below the state block.

This is a sequential refinement rather than a purely combinational identity. The formal harness first proves the joined logical relation between baseline state and optimized (C, p), then checks every public output on every future cycle:

$$\texttt{baseline.reg\_shake} = C \gg (8p) \quad \Longrightarrow \quad \texttt{outputs}_B = \texttt{outputs}_O.$$

> **Correctness result.** All 239 recorded cycle checks pass, and EQY proves the transition relation inductively after the declared reset. The PPA result is downstream of, and conditional on, this exact cycle-equivalence gate.

The conceptual step is not that the source is longer or the function larger. It is that physical state encoding changes while the externally visible sequential machine remains identical. That proof obligation distinguishes this case from a local Boolean rewrite or an arithmetic-width reduction.

9. PPA environment and statistics

OPEN 45 NM FPGA PROXY

The target matches the other portfolio cases: VTR's homogeneous `k6_N10_I40_Fi6_L4_frac0_ff1_45nm` architecture, with six-input LUTs clustered ten per logic block and the architecture's routing and timing model. Power uses PTM45 properties at 0.9 V and $85\,^{\circ}$C. MWTA is minimum-width-transistor area, not square-micrometre physical area.

ACE supplies a deterministic probabilistic activity model: primary-input static probability 0.5, switching probability 0.2, clock probability 0.2 and ACE seed 1. Baseline and optimized legs use the same placement seed and the same activity contract. This estimates relative active total power for the fixed academic model; it is not application-trace energy.

The primary per-pair score is

$$r_p = \sqrt[3]{r_{A,p}\, r_{D,p}\, r_{P,p}}, \qquad \widehat{r} = \exp\left(\frac{1}{64}\sum_{p=1}^{64} \log r_p\right),$$

where A is total MWTA, D is post-route critical-path delay and P is active total power. Fmax is the reciprocal of routed delay and is not a separate score term.

Two-sided 95% Student-t intervals use 63 degrees of freedom on paired log ratios. The one-sided lower improvements are 15.5895% area, 5.7243% delay, 4.3538% power and 9.3992% composite. All remain on the improvement side of zero under the predeclared rule.

Pair policy. Three search seeds and the earlier 12-pair confirmation were observed before the publication protocol. The optimized RTL hash was then frozen, and the disjoint seeds 101–164 were measured exactly once as the 64-pair publication pool. The pool was not extended, filtered or pooled with earlier evidence.

These intervals quantify implementation-seed variability for this exact target. They do not cover process, voltage, temperature, workload, side-channel leakage or tool-version uncertainty.

10. Every paired publication result

PRIMARY PPA DISTRIBUTIONS

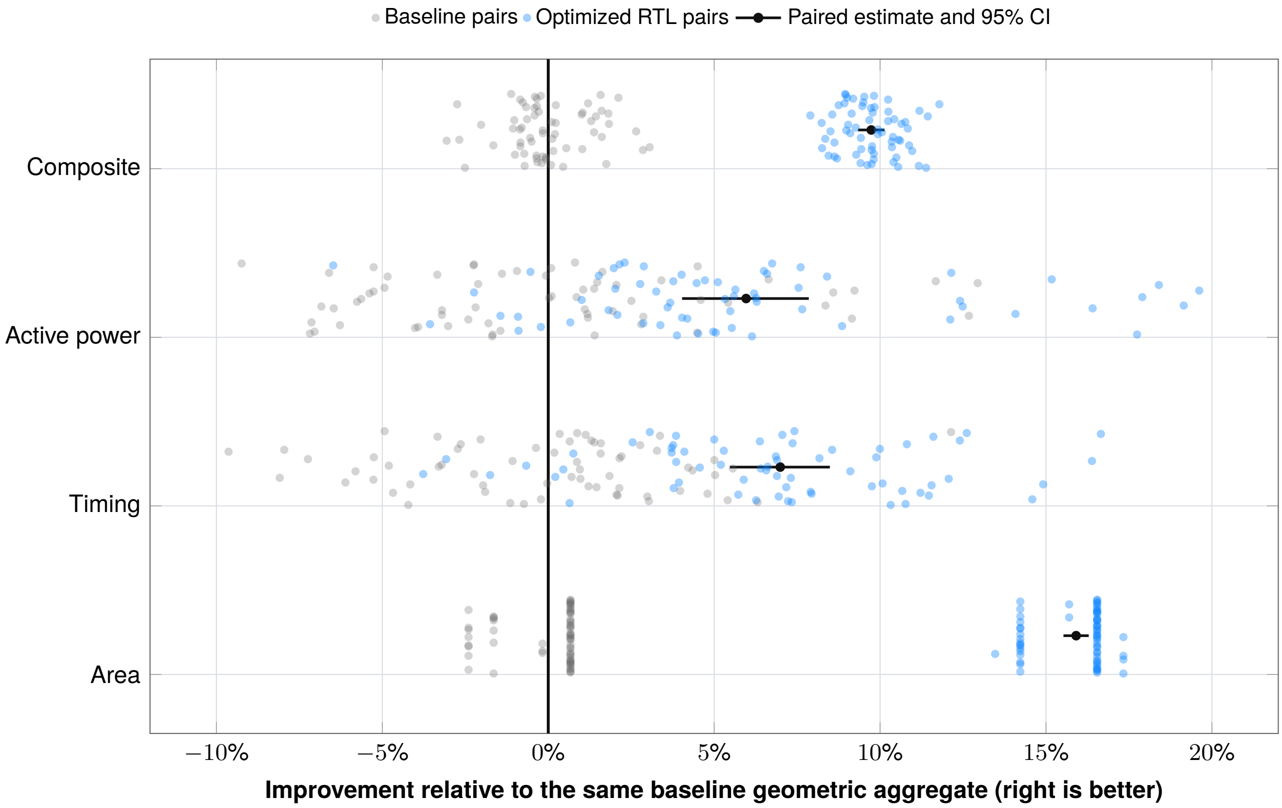


FIGURE 2 · OVERLAID BASELINE AND OPTIMIZED PAIR CLOUDS

Gray and blue points are absolute results normalized to the same baseline geometric aggregate. The same pair receives the same vertical jitter. Black points and segments show the paired estimator and interval that determine confidence.

Metric	Wins	Ties	Losses
Area	64	0	0
Timing	58	0	6
Active power	52	0	12
Composite	64	0	0

11. Secondary metrics and evidence identity

MEASURED CONSEQUENCES AND PUBLIC AUDIT TRAIL

Metric	Baseline aggregate	Optimized aggregate	Interpretation
Fmax	246.6840 MHz	265.2289 MHz	7.5177% increase; reciprocal of routed delay
Packed CLBs	337	270	19.8813% fewer in every pair
Used logic area	18,162,300 MWTA	14,551,400 MWTA	19.8813% lower in every pair
Routing area	4,583,859 MWTA	4,575,960 MWTA	Placement-seed dependent; included in total area
Initial confirmation	1.000000	0.905348	9.4652% on 12 earlier pairs; selection history only

The optimized netlist uses no multiplier or memory hard blocks under this academic architecture. CLB and used-logic-area reductions are stable structural consequences; routing, timing and power vary with placement. None of these values is inferred from RTL line count or an AST proxy.

Public evidence object	SHA-256 / identity
Compact certification record	`3d8b6120db2957da6c4d3cd4569e73df180ba07d5c624c9ae4bc1f4d2d37406d`
Publication protocol	`cce84fcc54e048137b5a3a89bb0a4e3b92141106ec8330bccb344812b0b02a2c`
Evaluator	`b91dc19ab42285d4477d6ca9f1d97c9216bb9725e114026a8c123e71713c1106`
Baseline publication record	`a919d24bc01948d94510118e4dd7a8271c97841ab4db1ec3327cf9aab4ba24e6`
Optimized publication record	`1e6975a17af1880629d5b5f9c203c71e346751bd951d5475b867efcd789e10fd`
VTR architecture	`262792caef81931ae09b77d252158bde03c78954175d4b761e6d437317575307`
PTM technology	`3080dea13bf7134109f8a83c79426ec0965e07639a61866fe9ae4bd05b5227cb`
Full raw-evidence archive	`mlkem-cbd-vtr45-full-evidence-v1.tar.gz`
Full archive SHA-256	`bc628858c80725a373df8ded0155d7bbbbd71e4aef84038f73aa1a19c290598f`

The 541-member archive binds the source, protocol, verifier inputs, correctness logs and all baseline/optimized VTR and ACE reports for publication seeds 101–164. Its optimized record differs from the internal source record only by a declared sanitization of the local evidence-root path.

12. Reproduction and verifier trust model

WHAT EACH PUBLIC COMMAND ESTABLISHES

The repository separates four levels of evidence:

1. **Git identity**: checksums cover the exact compact package and the commit records its history.
2. **Compact consistency**: `python3 verify.py` validates strict schemas, RTL hashes, exact patch, 64 paired rows, formulas, intervals, authority and acceptance decision.
3. **Raw provenance replay**: `python3 verify.py --evidence-archive ...` verifies all archive members, correctness logs and metrics re-extracted from 128 timing and ACE power reports.
4. **Fresh EDA reproduction**: reruns the published contract with the pinned tools and target; this is a separate, stronger review step and is not performed by the compact verifier.

```
git clone https://github.com/juan-fernandez-gotherlabs/\
rtl-optimization-case-study.git
cd rtl-optimization-case-study/cases/mlkem-cbd

python3 verify.py
python3 verify.py --evidence-archive \
  mlkem-cbd-vtr45-full-evidence-v1.tar.gz
python3 -m unittest discover -s tests -v
```

The verifier uses explicit checks rather than Python `assert`, requires exact manifest coverage, rejects duplicate or unexpected fields, non-finite measurements, unsafe archive paths, altered RTL, patch drift, authority changes, metric substitution, score substitution, incomplete evidence and any raw value that differs from the compact certificate. Its adversarial unit suite exercises these cases in CI.

A fresh EDA rerun requires Linux/amd64, the pinned VTR commit, exact architecture and PTM files, the ACE activity transform, the EQY formal configuration and the frozen publication protocol. A different target or activity model is a new experiment.

> **Trust boundary.** The Git commit and release identify the published package. Raw-provenance verification re-derives the reported result from recorded files, but it is not a fresh third-party tool execution and cannot establish physical-chip behaviour. “Certification” here means project-contract acceptance, not accredited assurance.

13. Limits and customer transfer

WHAT THE RESULT DOES AND DOES NOT ESTABLISH

The portfolio now covers three distinct optimization classes:

Case	Domain	Transformation class
SHA-1	Sequential legacy cryptography	Shared Boolean structure and direct state readout
INT8 MatVec	Quantized AI arithmetic	Range-correct widths and balanced accumulation
ML-KEM CBD	Post-quantum cryptographic IP	Alternative physical state representation with a formal refinement relation

Cross-case percentages are not a cross-circuit ranking. The new case is not larger by source line count; it is conceptually different because it changes the encoding and evolution of sequential state while proving the same public machine. Its industrial relevance is post-quantum cryptographic hardware, particularly teams evaluating FPGA or ASIC ML-KEM IP.

This report does not provide:

- ASIC synthesis, place-and-route, PVT analysis or signoff;
- commercial-FPGA implementation data, Vivado, Quartus or board measurements;
- measured power, measured energy or manufactured-silicon evidence;
- side-channel resistance, masking quality, fault resistance or security certification;
- certification of HOPE-MLKEM or of a complete FIPS 203 implementation;
- proof that the same representation improves another technology, integration or workload;
- exact PPA attribution to one line or one structural view of the accepted patch.

> **Transfer to a customer pilot.** Replace the academic proxy with customer-owned RTL, security assumptions, libraries, constraints, activity, tools and signoff policy. The transferable capability is a small inspectable state transformation evaluated by a frozen contract and justified with cycle regression, sequential formal proof and paired implementation evidence.

Primary references

- NIST FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard
- HWSec-CSIC HOPE-MLKEM source repository
- VTR/VPR documentation and power estimation
- Public case-study repository
- Göther Labs

> **Conclusion.** Within the declared VTR/PTM45 LUT6 proxy and exact sequential contract, the optimized CBD state representation passes the recorded cycle and formal gates and reduces the prospectively confirmed paired primary PPA estimator by 9.7338% (95% CI 9.3331% to 10.1327%). The before/after RTL, compact calculations and every raw publication-pair metric are publicly auditable.